# NIWA

# POST CYCLONE GABRIELLE HAWKES BAY LiDAR

# VOLUME: PRJ44793_02-04

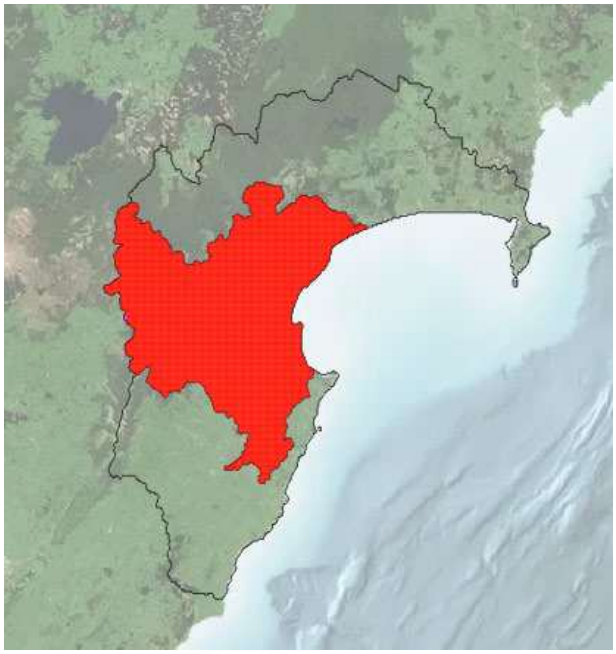
## PROJECT SUMMARY

The National Institute of Water and Atmospheric Research Limited (NIWA) contracted Woolpert to capture and process LiDAR and concurrent imagery over Hawkes Bay, post Cyclone Gabrielle.

This report is for the products supplied for Priority One in volumes – PRJ44793_02, PRJ44793_03 & PRJ44793_04, approximately 5,519 $km^2$ in total. Resubmissions following LINZ QA were completed in May 2024.

The majority of this area was captured between the 20th of September 2023 – 21st of December 2023, and a small refly on the western edge was captured on 23 April 2024.

The survey was planned to achieve ≤ 20cm vertical accuracy (95% CI), ≤ 100cm horizontal accuracy (95% CI) with an emitted pulse density of 8ppsm, and ground classification to ICSM level 2.



# DATA SUMMARY

This volume includes the following data in NZTM2000 projection, NZVD2016 vertical datum:

- 15,970 x NZTopo50 1:1000 tiles are included in Hawkes Bay Priority One
- 15,970 tiles of ICSM L2 Colourised Classified Point Cloud data in LAZ v1.4 format
- 15,964 tiles of 1m cell DEM in GeoTIFF format
- 15,964 tiles of 1m cell DSM in GeoTIFF format
- 15,964 tiles of 1m cell CHM in GeoTIFF format
- 15,970 tiles of 12.5cm gsd RGBI Imagery in GeoTIFF format
- Ancillary files in ESRI Shapefile format – Tile Index, Delivery Extent, Flightlines, Hydro Breaklines.
- File listing in text file format
- Metadata file: This document in PDF format

# CONTENTS

1. Data Information ..... 4
2. Metadata ..... 6
3. Conditions Of Supply ..... 10
4. Validation ..... 11

# 1. DATA INFORMATION

Data supply: LINZ AWS
Number of files: 130,416 data files, 1 file list, 1 survey report, 1 metadata file
Data formatted on: 20/03/2024 - 09/04/2024, 14-22/05/2024
Metadata Document: This file

| Previous Deliveries | Date | Title | Contents |
|---|---|---|---|
| PRJ44793_01 | 22.12.2023 | Post Cyclone Gabrielle Hawkes Bay LiDAR | Sample of deliverables from P1-SE |
| PRJ44793_02 | 08.04.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P1-SE |
| PRJ44793_03 | 05.04.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P1-N |
| PRJ44793_04 | 06.04.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P1-W |
| PRJ44793_04-02 | 14-22.05.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | Resubmissions for Priority 1 blocks SE, N, W |

| File Naming in this Delivery | Contents |
|---|---|
| CL2_BH36_2023_1000_3441.laz | Colourised classified point cloud in LAZ v1.4 format |
| DEM_BH36_2023_1000_3441.tif/tfw | 1m grid DEM |
| DSM_BH36_2023_1000_3441.tif/tfw | 1m grid DSM |
| CHM_BH36_2023_1000_3441.tif/tfw | 1m grid CHM (canopy height above ground) |
| RGBI_BH36_2023_1000_3441.tif/tfw | 12.5cm gsd RGBI imagery |

| | |
|---|---|
| Ancillary files:<br>PRJ44793_HB_P1-SE_HydroBreaklines_NZTM<br>PRJ44793_HB_P1-SE_LiDAR_Extent_NZTM<br>PRJ44793_HB_P1-SE_OceanClip_NZTM<br>PRJ44793_HB_P1-SE_TileIndex_NZTM<br>PRJ44793_HB_P1-SE_Trajectories_NZTM | ESRI Shapefile format:<br>Volume 02 Hydro breaklines<br>Volume 02 Extent<br>Volume 02 Ocean clip boundary<br>Volume 02 Tile Index<br>Volume 02 Flight lines |
| Ancillary files:<br>PRJ44793_HB_P1-N_HydroBreaklines_NZTM<br>PRJ44793_HB_P1-N_LiDAR_Extent_NZTM<br>PRJ44793_HB_P1-N_OceanClip_NZTM<br>PRJ44793_HB_P1-N_TileIndex_NZTM<br>PRJ44793_HB_P1-N_Trajectories_NZTM | ESRI Shapefile format:<br>Volume 03 Hydro breaklines<br>Volume 03 Extent<br>Volume 03 Ocean clip boundary<br>Volume 03 Tile Index<br>Volume 03 Flight lines |
| Ancillary files:<br>PRJ44793_HB_P1-W_HydroBreaklines_NZTM<br>PRJ44793_HB_P1-W_LiDAR_Extent_NZTM<br>PRJ44793_HB_P1-W_TileIndex_NZTM<br>PRJ44793_HB_P1-W_Trajectories_NZTM | ESRI Shapefile format:<br>Volume 04 Hydro breaklines<br>Volume 04 Extent<br>Volume 04 Tile Index<br>Volume 04 Flight lines |
| Readme_PRJ44793_02-04.pdf | Metadata Report |
| PRJ44793_02-04_HawkesBay_P1_LiDAR_Report.pdf | Survey Report |
| PRJ44793_02-04_FileListing.txt | List of product files delivered in this volume |

# 2. METADATA

| Source Data | Source | Description | Ref No | Date |
|---|---|---|---|---|
| LiDAR & Imagery | Woolpert | TerrainMapper2 - 527 | FL020423 | 20.09.2023 |
| | | TerrainMapper2 – 527 | FL020424 | 21.09.2023 |
| | | TerrainMapper2 – 527 | FL020430 | 21.09.2023 |
| | | TerrainMapper2 – 527 | FL020511 | 01.10.2023 |
| | | TerrainMapper2 – 527 | FL020533 | 04.10.2023 |
| | | TerrainMapper2 – 527 | FL020539 | 04.10.2023 |
| | | TerrainMapper2 – 527 | FL020542 | 05.10.2023 |
| | | TerrainMapper2 – 527 | FL020547 | 05.10.2023 |
| | | TerrainMapper2 – 527 | FL020601 | 11.10.2023 |
| | | TerrainMapper2 – 527 | FL020608 | 13.10.2023 |
| | | TerrainMapper2 – 527 | FL020623 | 15.10.2023 |
| | | TerrainMapper2 – 527 | FL020650 | 19.10.2023 |
| | | TerrainMapper2 – 527 | FL020655 | 20.10.2023 |
| | | TerrainMapper2 – 527 | FL020842 | 12.11.2023 |
| | | TerrainMapper2 – 527 | FL020875 | 14.11.2023 |
| | | TerrainMapper2 – 527 | FL021135 | 15.12.2023 |
| | | TerrainMapper2 – 527 | FL021141 | 16.12.2023 |
| | | TerrainMapper2 – 527 | FL021156 | 17.12.2023 |
| | | TerrainMapper2 – 527 | FL021159 | 17.12.2023 |
| | | TerrainMapper2 – 527 | FL021182 | 21.12.2023 |
| | | TerrainMapper2 – 527 | FL022290 | 23.04.2024 |
| GPS Base Data | GeoNET /LINZ | GeoNET CORS, LINZ GDB co-ordinates | As above | As above |
| Control | Sounds Surveying | RTK GNSS | PRJ44793 | 27/09/2023 – 8/10/2023 |

| LiDAR Characteristics | Description |
|---|---|

| | |
|---|---|
| Format | LAZ 1.4 |
| Emitted Density | 8 ppm2 |
| Tile size | 480m x 720m (NZ Topo 50, 1:1000 tiles) |
| ICSM Classification | Level 2. Ground surface improvement |
| Capture Constraints | Napier Estuary and the coast line between Esk and Ngaruroro rivers captured within 1.5 hours of low tide. Remaining coastline captured within 3 hours of low tide. |

| Number | Point Class | Description | ICSM | CI % |
|---|---|---|---|---|
| 1 | Default | Unclassified | 1 | 95 |
| 2 | Ground | Ground | 2 | 98 |
| 3 | Low vegetation | < 2.0 m | 1 | 95 |
| 4 | Medium vegetation | 2.0 – 8.0 m | 1 | 95 |
| 5 | High vegetation | > 8.0 m | 1 | 95 |
| 6 | Buildings, structures | Buildings, houses, sheds, silos etc. | 1 | 95 |
| 7 | Low Noise | Spurious low point returns (unusable) | 2 | 98 |
| 9 | Water | Any point in water | 2 | 98 |
| 17 | Bridge | Any bridge or overpass | 2 | 98 |
| 18 | High Noise | Spurious high point returns (unusable) | 2 | 98 |

| Ortho Characteristics | Description |
|---|---|
| Format | GeoTIFF & ECW |
| Ground Sample Distance | 12.5cm |
| Terrain Model | Concurrent LiDAR |
| Tile size | 480m x 720m (NZ Topo 50, 1:1000 tiles) |
| Sample Type | 8-bit Integer |
| Image Bands | RGBI |
| Orientation/AT | Aero Triangulation/GPS-IMU |
| Capture Constraints | Nil specific to imagery – captured under LiDAR capture conditions above, in daylight hours |

| Reference Systems | Horizontal | Vertical |
|---|---|---|
| Datum | NZGD2000 | NZVD2016 |
| Projection | NZTM2000 | N/A |
| Geoid Model | N/A | NZGeoid2016 |

| Accuracy Specification | Measured Point | Derived Point | Basis of Estimation |
|---|---|---|---|
| Control Points | 0.05m | | Survey Methodology |
| LiDAR (Horizontal) | 0.40m | | Project design |
| LiDAR (Vertical) | 0.10m | | Project design |
| Orthophoto | 0.30m | | Project design |

## Notes On Expected Accuracy

- Values shown represent standard error (68% confidence level or 1 sigma), in metres
- “Derived points” are those interpolated from a terrain model.
- “Measured points” are those observed directly.
- Accuracy estimates for terrain modeling by LiDAR refer to the terrain definition on clear ground.
- LASER strikes have been classified into “ground” and “non-ground”, based upon algorithms tailored for major terrain/vegetation combinations existing in the project area. The definition of the ground may be less accurate in isolated pockets of dissimilar terrain/vegetation combinations.

## Limitations of Data

- The definition of the ground under trees may be less accurate.
- Buildings and vegetation have been classified automatically: ISCM Level 1, 95% confidence. These classes may contain artefacts such as powerlines, car’s, containers etc. Be aware when using the LAZ and CHM products.

## Data Validation – LiDAR Data

- Vertical Accuracy Validation - Ground data in this volume has been compared to ~2800 test points obtained by field survey and assumed to be error-free. The test points were distributed in 62 groups across the mapping area and located on clear ground. Comparison of the test points with elevations interpolated from measured data resulted in:

  Mean difference: 0.006 m

  Std Deviation: 0.034 m

  Standard Error (RMS): 0.035 m or 0.068 m @ 95% CI

  DEM comparisons were made between adjacent blocks to ensure consistency, blocks were found to align within 0.023m.

  A shift of -0.127 was applied to the data before testing against the check points to produce the results shown above.

- Horizontal Accuracy – the LiDAR data was compared to visible horizontal control using the intensity imagery. The data was found to fit well. The expected horizontal accuracy at 95% CI is 38cm, within the specification of 100cm.
- Data classification has been manually checked and edited against available imagery.
- More information is available in the Survey Report: PRJ44793_02-04_HawkesBay_P1_LiDAR_Report.pdf

## Data Validation – Orthophoto Image

- This data has not been field tested for accuracy. Spot checks comparing the orthophoto with horizontal check points supported the estimated horizontal accuracy of < 0.300 m. Full proof of accuracy achieved requires comparison to independent test points.

# 3. CONDITIONS OF SUPPLY

The data in this volume has been commissioned by the National Institute of water and Atmospheric Research Limited **NIWA**.

The data in this volume is provided by Woolpert to **NIWA** under the Terms of Engagement described in the **NIWA Woolpert Imagery Contract Agreement, dated 16th August 2023**. Which provides the client, **NIWA** (or its nominee) with ownership of the deliverables to fully protect, realise and use all delivered data and reports, allowing release of data for re-use with a Creative Commons license (CC BY) with attribution to the buyer in line with the New Zealand Government Open Access Licensing framework (NZGOAL). This specification places no restrictions on the rights of Woolpert to resell data or derivative products.

1. This file (Readme_PRJ44793_02-04_rev12wq.pdf) will always be stored with the unaltered data contained in this volume.

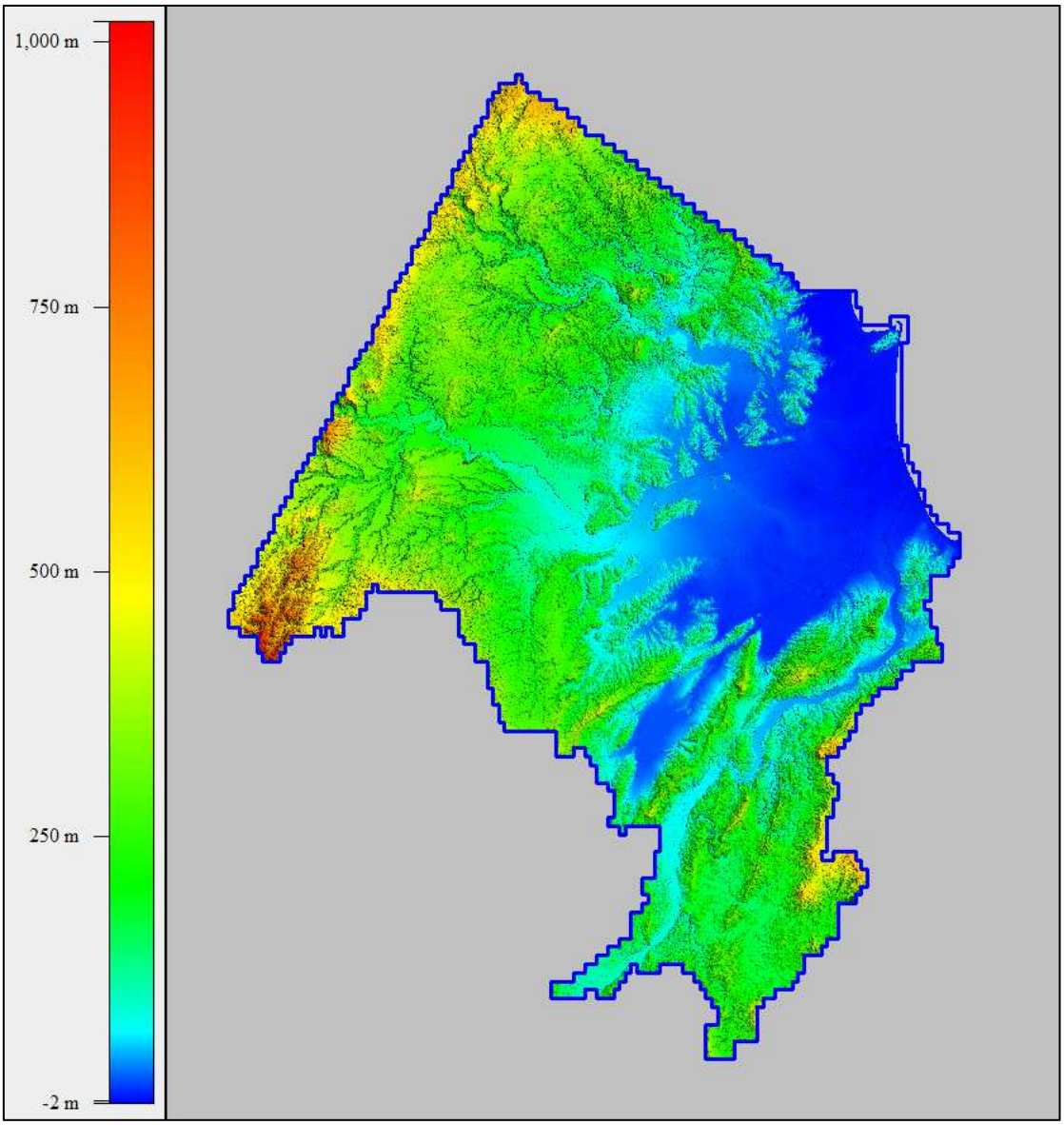
This data is provided in accordance with the specifications agreed with NIWA and Land Information NZ. Any problems associated with the information in the data files contained in this volume should be reported to Woolpert, Asia-Pacific.

Woolpert NZ Limited
Level 1
6 Ossian St
Napier 4110
New Zealand

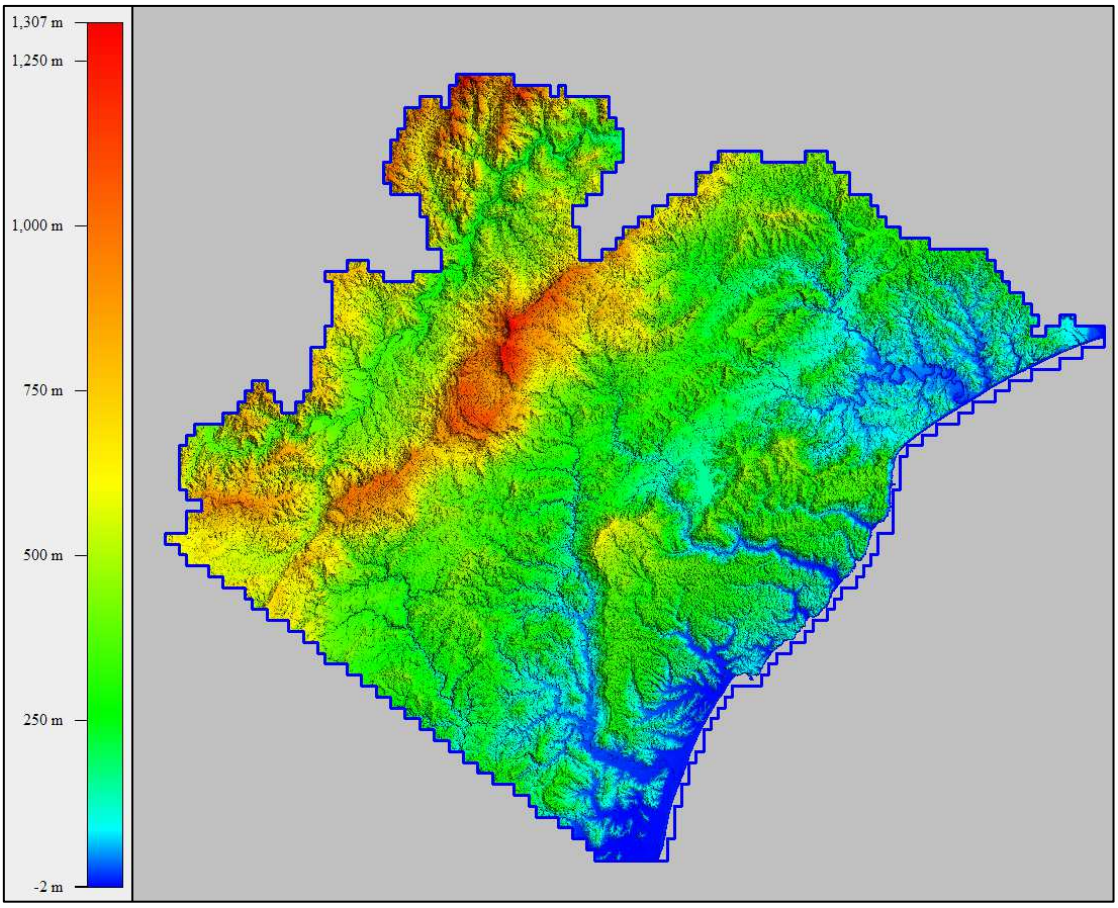
woolpert.com
Twitter Facebook LinkedIn Instagram

# 4. VALIDATION

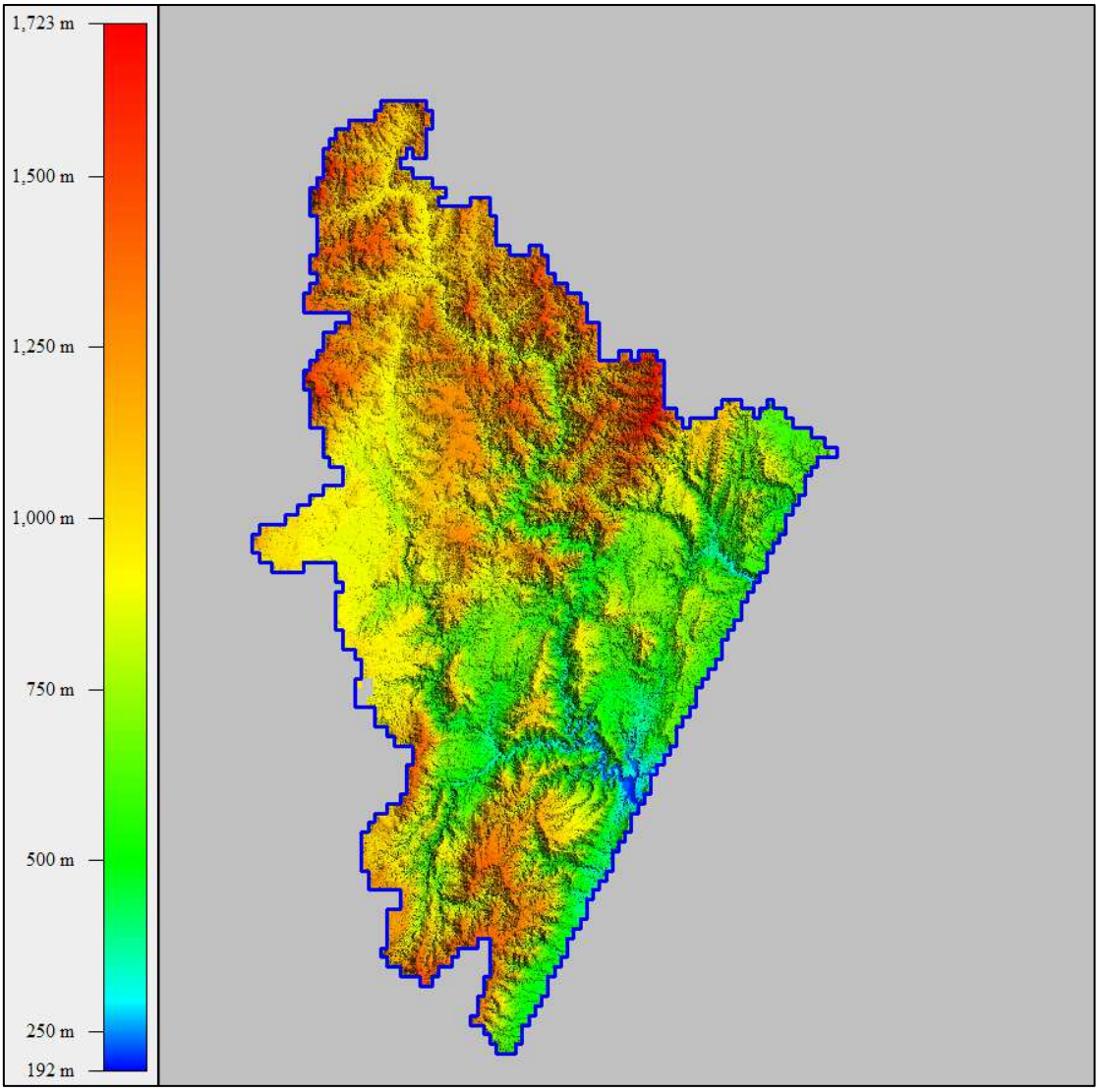
PRJ44793_02 – P1-SE



PRJ44793_03 – P1-N



PRJ44793_04 – P1-W



# NIWA

# POST CYCLONE GABRIELLE HAWKES BAY LiDAR

# VOLUME: PRJ44793_05

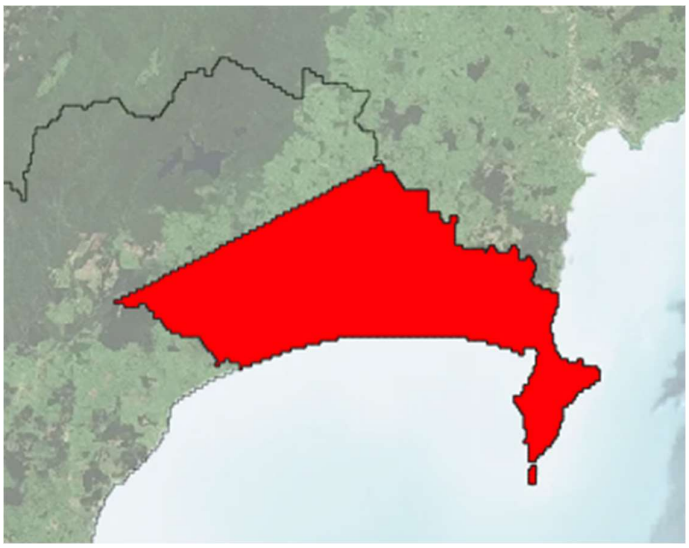
## PROJECT SUMMARY

The National Institute of Water and Atmospheric Research Limited (NIWA) contracted Woolpert to capture and process LiDAR and concurrent imagery over Hawkes Bay, post Cyclone Gabrielle.

This report is for the products supplied for Priority Two in volume – PRJ44793_05, approximately 2,015 $km^2$. Resubmissions following LINZ QA were completed in August.

This area was captured between the 20th of September 2023 – 17th of December 2023.

The survey was planned to achieve ≤ 20cm vertical accuracy (95% CI), ≤ 100cm horizontal accuracy (95% CI) with an emitted pulse density of 8ppsm, and ground classification to ICSM level 2.



# DATA SUMMARY

This volume includes the following data in NZTM2000 projection, NZVD2016 vertical datum:

- 5,831 x NZTopo50 1:1000 tiles are included in PRJ44793_05
- 5,831 tiles of ICSM L2 Colourised Classified Point Cloud data in LAZ v1.4 format
- 5,814 tiles of 1m cell DEM in GeoTIFF format
- 5,814 tiles of 1m cell DSM in GeoTIFF format
- 5,814 tiles of 1m cell CHM in GeoTIFF format
- 5,831 tiles of 12.5cm gsd RGBI Imagery in GeoTIFF format
- Ancillary files in ESRI Shapefile format – Tile Index, Delivery Extent, Flightlines, Hydro Breaklines.
- File listing in text file format
- Metadata file: This document in PDF format

# CONTENTS

1. Data Information ..... 4
2. Metadata ..... 5
3. Conditions Of Supply ..... 9
4. Validation ..... 10

# 1. DATA INFORMATION

Data supply: LINZ AWS
Number of files: 52,397 data files, 1 file list, 1 survey report, 1 metadata file
Data formatted on: 12-28/05/2024. Resubmissions: 13 & 23/08/2024
Metadata Document: This file

| Previous Deliveries | Date | Title | Contents |
|---|---|---|---|
| PRJ44793_01 | 22.12.2023 | Post Cyclone Gabrielle Hawkes Bay LiDAR | Sample of deliverables from P1-SE |
| PRJ44793_02 | 08.04.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P1-SE |
| PRJ44793_03 | 05.04.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P1-N |
| PRJ44793_04 | 06.04.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P1-W |
| PRJ44793_04-02 | 14-22.05.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | Resubmissions for Priority 1 blocks SE, N, W |
| PRJ44793_05 | 12-28/05/2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P2-NE |

| File Naming in this Delivery | Contents |
|---|---|
| CL2_BG41_2023_1000_4237.laz | Colourised classified point cloud in LAZ v1.4 format |
| DEM_BG41_2023_1000_4237.tif/tfw | 1m grid DEM |
| DSM_BG41_2023_1000_4237.tif/tfw | 1m grid DSM |
| CHM_BG41_2023_1000_4237.tif/tfw | 1m grid CHM (canopy height above ground) |
| RGBI_BG41_2023_1000_4237.tif/tfw | 12.5cm gsd RGBI imagery |

| | |
|---|---|
| Ancillary files:<br>PRJ44793_HB_P2-NE_HydroBreaklines_NZTM<br>PRJ44793_HB_P2-NE_LiDAR_Extent_NZTM<br>PRJ44793_HB_P2-NE_OceanClip_NZTM<br>PRJ44793_HB_P2-NE_TileIndex_NZTM<br>PRJ44793_HB_P2-NE_Trajectories_NZTM | ESRI Shapefile format:<br>Volume 05 Hydro breaklines<br>Volume 05 Extent<br>Volume 05 Ocean clip boundary<br>Volume 05 Tile Index<br>Volume 05 Flight lines |
| Readme_PRJ44793_05.pdf | Metadata Report |
| PRJ44793_05_HawkesBay_LiDAR_Report.pdf | Survey Report |
| PRJ44793_05_FileListing.txt | List of product files delivered in this volume |

# 2. METADATA

| Source Data | Source | Description | Ref No | Date |
|---|---|---|---|---|
| LiDAR & Imagery | Woolpert | TerrainMapper2 - 527 | FL020423 | 20.09.2023 |
| | | TerrainMapper2 – 527 | FL020601 | 12.10.2023 |
| | | TerrainMapper2 – 527 | FL020617 | 14.10.2023 |
| | | TerrainMapper2 – 527 | FL020618 | 14.10.2023 |
| | | TerrainMapper2 – 527 | FL020623 | 16.10.2023 |
| | | TerrainMapper2 – 527 | FL020650 | 19.10.2023 |
| | | TerrainMapper2 – 527 | FL020655 | 20.10.2023 |
| | | TerrainMapper2 – 527 | FL020889 | 16.11.2023 |
| | | TerrainMapper2 – 527 | FL020875 | 14.11.2023 |
| | | TerrainMapper2 – 527 | FL021060 | 07.12.2023 |
| | | TerrainMapper2 – 527 | FL021081 | 09.12.2023 |
| | | TerrainMapper2 – 527 | FL021150 | 16.12.2023 |
| | | TerrainMapper2 – 527 | FL021156 | 17.12.2023 |
| | | TerrainMapper2 – 527 | FL021159 | 17.12.2023 |
| GPS Base Data | GeoNET /LINZ | GeoNET CORS, LINZ GDB co-ordinates | As above | As above |
| Control | Sounds Surveying | RTK GNSS | PRJ44793 | 27/09/2023 – 8/10/2023 & 23-28/11/2023 |

| LiDAR Characteristics | Description |
|---|---|
| Format | LAZ 1.4 |
| Emitted Density | 8 ppm2 |
| Tile size | 480m x 720m (NZ Topo 50, 1:1000 tiles) |
| ICSM Classification | Level 2. Ground surface improvement |
| Capture Constraints | Coastline captured within 3 hours of low tide. |

| Number | Point Class | Description | ICSM | CI % |
|---|---|---|---|---|
| 1 | Default | Unclassified | 1 | 95 |
| 2 | Ground | Ground | 2 | 98 |
| 3 | Low vegetation | < 2.0 m | 1 | 95 |
| 4 | Medium vegetation | 2.0 – 8.0 m | 1 | 95 |
| 5 | High vegetation | > 8.0 m | 1 | 95 |
| 6 | Buildings, structures | Buildings, houses, sheds, silos etc. | 1 | 95 |
| 7 | Low Noise | Spurious low point returns (unusable) | 2 | 98 |
| 9 | Water | Any point in water | 2 | 98 |
| 17 | Bridge | Any bridge or overpass | 2 | 98 |
| 18 | High Noise | Spurious high point returns (unusable) | 2 | 98 |

| Ortho Characteristics | Description |
|---|---|
| Format | GeoTIFF |
| Ground Sample Distance | 12.5cm |
| Terrain Model | Concurrent LiDAR |
| Tile size | 480m x 720m (NZ Topo 50, 1:1000 tiles) |
| Sample Type | 8-bit Integer |
| Image Bands | RGBI |
| Orientation/AT | Aero Triangulation/GPS-IMU |
| Capture Constraints | Nil specific to imagery – captured under LiDAR capture conditions above, in daylight hours |

| Reference Systems | Horizontal | Vertical |
|---|---|---|
| Datum | NZGD2000 | NZVD2016 |
| Projection | NZTM2000 | N/A |
| Geoid Model | N/A | NZGeoid2016 |

| Accuracy Specification | Measured Point | Derived Point | Basis of Estimation |
|---|---|---|---|
| Control Points | 0.05m | | Survey Methodology |
| LiDAR (Horizontal) | 0.40m | | Project design |
| LiDAR (Vertical) | 0.10m | | Project design |
| Orthophoto | 0.30m | | Project design |

## Notes On Expected Accuracy

- Values shown represent standard error (68% confidence level or 1 sigma), in metres
- “Derived points” are those interpolated from a terrain model.
- “Measured points” are those observed directly.
- Accuracy estimates for terrain modeling by LiDAR refer to the terrain definition on clear ground.
- LASER strikes have been classified into “ground” and “non-ground”, based upon algorithms tailored for major terrain/vegetation combinations existing in the project area. The definition of the ground may be less accurate in isolated pockets of dissimilar terrain/vegetation combinations.

## Limitations of Data

- The definition of the ground under trees may be less accurate.
- Buildings and vegetation have been classified automatically: ISCM Level 1, 95% confidence. These classes may contain artefacts such as powerlines, car’s, containers etc. Be aware when using the LAZ and CHM products.

## Data Validation – LiDAR Data

- Vertical Accuracy Validation - Ground data in this volume has been compared to ~1270 test points obtained by field survey and assumed to be error-free. The test points were distributed in 23 groups across the mapping area and located on clear ground. Comparison of the test points with elevations interpolated from measured data resulted in:

  Mean difference: -0.007 m

  Std Deviation: 0.045 m

  Standard Error (RMS): 0.046 m or 0.090 m @ 95% CI

  DEM and control comparisons were made to the adjacent Priority One block to ensure consistency, these align within 0.01m.

  A shift of -0.110 was applied to the data before testing against the check points to produce the results shown above.

- Horizontal Accuracy – the LiDAR data was compared to visible horizontal control using the intensity imagery. The data was found to fit well. The expected horizontal accuracy at 95% CI is 38cm, within the specification of 100cm.
- Data classification has been manually checked and edited against available imagery.
- More information is available in the Survey Report: PRJ44793_05-07_HawkesBay_P2_LiDAR_Report.pdf

## Data Validation – Orthophoto Image

- This data has not been field tested for accuracy. Spot checks comparing the orthophoto with horizontal check points supported the estimated horizontal accuracy of < 0.300 m. Full proof of accuracy achieved requires comparison to independent test points.

# 3. CONDITIONS OF SUPPLY

The data in this volume has been commissioned by the National Institute of water and Atmospheric Research Limited **NIWA**.

The data in this volume is provided by Woolpert to **NIWA** under the Terms of Engagement described in the **NIWA Woolpert Imagery Contract Agreement, dated 16th August 2023**. Which provides the client, **NIWA** (or its nominee) with ownership of the deliverables to fully protect, realise and use all delivered data and reports, allowing release of data for re-use with a Creative Commons license (CC BY) with attribution to the buyer in line with the New Zealand Government Open Access Licensing framework (NZGOAL). This specification places no restrictions on the rights of Woolpert to resell data or derivative products.

1. This file (Readme_PRJ44793_05.pdf) will always be stored with the unaltered data contained in this volume.

This data is provided in accordance with the specifications agreed with NIWA and Land Information NZ. Any problems associated with the information in the data files contained in this volume should be reported to Woolpert, Asia-Pacific.

Woolpert NZ Limited
Level 1
6 Ossian St
Napier 4110
New Zealand

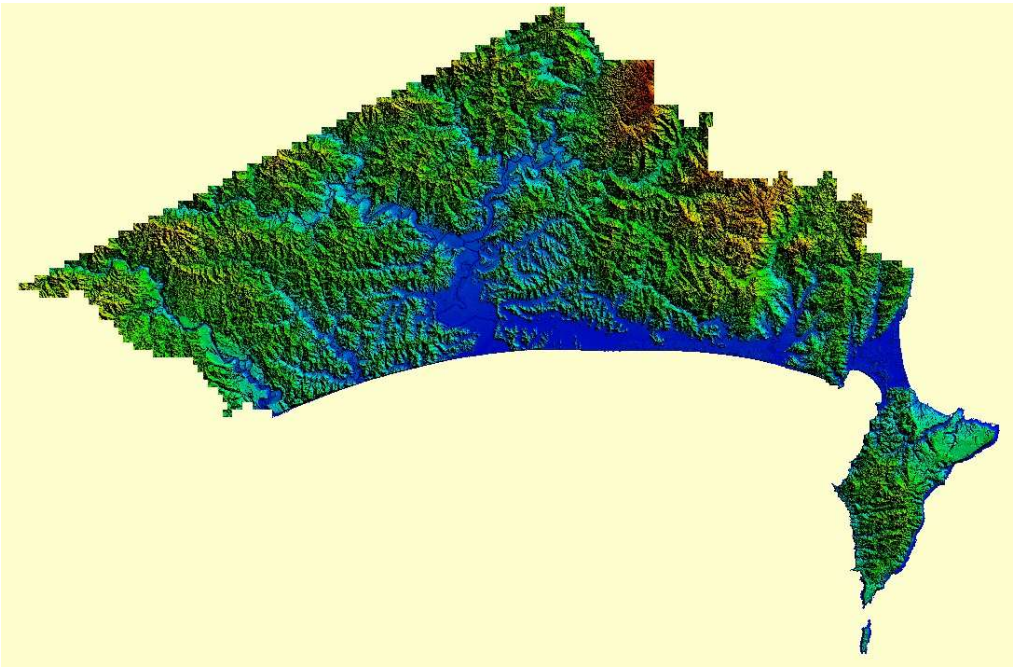
woolpert.com
Twitter Facebook LinkedIn Instagram

# 4. VALIDATION

PRJ44793_05 – P2-NE

Imagery

Colour Elevation



# NIWA

# POST CYCLONE GABRIELLE HAWKES BAY LiDAR

# VOLUME: PRJ44793_06

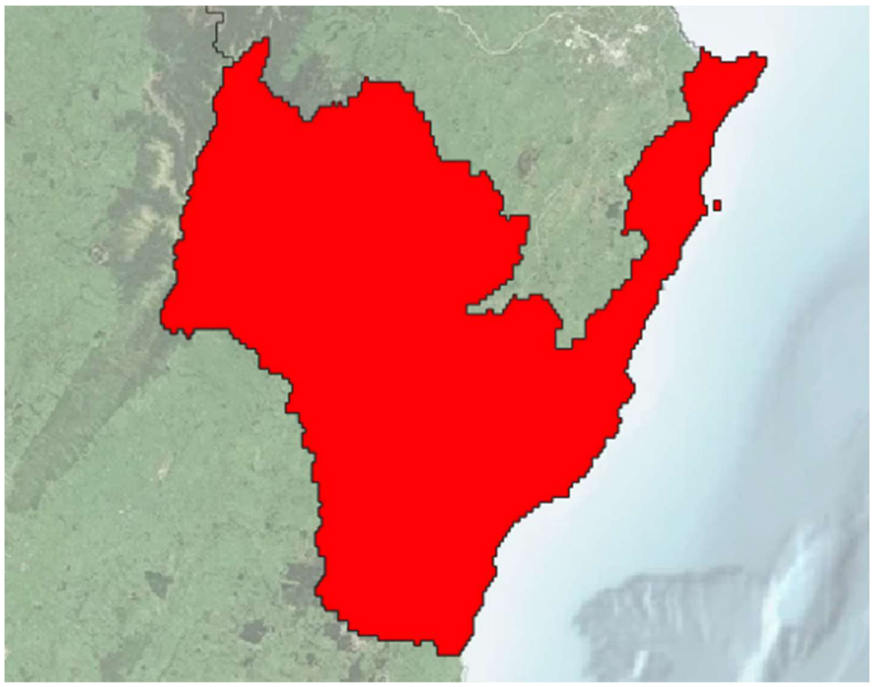
## PROJECT SUMMARY

The National Institute of Water and Atmospheric Research Limited (NIWA) contracted Woolpert to capture and process LiDAR and concurrent imagery over Hawkes Bay, post Cyclone Gabrielle.

This report is for the products supplied for Priority Two in volume – PRJ44793_06, approximately 3,548 $km^2$. Resubmission following LINZ QA were completed in August.

This area was captured between the 20th of September 2023 – 19th of April 2024.

The survey was planned to achieve ≤ 20cm vertical accuracy (95% CI), ≤ 100cm horizontal accuracy (95% CI) with an emitted pulse density of 8ppsm, and ground classification to ICSM level 2.



# DATA SUMMARY

This volume includes the following data in NZTM2000 projection, NZVD2016 vertical datum:

- 10,267 x NZTopo50 1:1000 tiles are included in PRJ44793_06
- 10,267 tiles of ICSM L2 Colourised Classified Point Cloud data in LAZ v1.4 format
- 10,235 tiles of 1m cell DEM in GeoTIFF format
- 10,235 tiles of 1m cell DSM in GeoTIFF format
- 10,235 tiles of 1m cell CHM in GeoTIFF format
- 10,267 tiles of 12.5cm gsd RGBI Imagery in GeoTIFF format
- Ancillary files in ESRI Shapefile format – Tile Index, Delivery Extent, Flightlines, Hydro Breaklines.
- File listing in text file format
- Metadata file: This document in PDF format

# CONTENTS

1. Data Information ..... 4
2. Metadata ..... 5
3. Conditions Of Supply ..... 9
4. Validation ..... 10

# 1. DATA INFORMATION

Data supply: LINZ AWS
Number of files: 92,231 data files, 1 file list, 1 survey report, 1 metadata file
Data formatted on: 19/06/2024 – 17/07/2024, resubmissions 30/08/2024
Metadata Document: This file

| Previous Deliveries | Date | Title | Contents |
|---|---|---|---|
| PRJ44793_01 | 22.12.2023 | Post Cyclone Gabrielle Hawkes Bay LiDAR | Sample of deliverables from P1-SE |
| PRJ44793_02 | 08.04.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P1-SE |
| PRJ44793_03 | 05.04.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P1-N |
| PRJ44793_04 | 06.04.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P1-W |
| PRJ44793_04-02 | 14-22.05.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | Resubmissions for Priority 1 blocks SE, N, W |
| PRJ44793_05 | 12-28/05/2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P2-NE |

| File Naming in this Delivery | Contents |
|---|---|
| CL2_BM38_2023_1000_4509.laz | Colourised classified point cloud in LAZ v1.4 format |
| DEM_ BM38_2023_1000_4509.tif/tfw | 1m grid DEM |
| DSM_ BM38_2023_1000_4509.tif/tfw | 1m grid DSM |
| CHM_ BM38_2023_1000_4509.tif/tfw | 1m grid CHM (canopy height above ground) |
| RGBI_ BM38_2023_1000_4509.tif/tfw | 12.5cm gsd RGBI imagery |

| | |
|---|---|
| Ancillary files:<br>PRJ44793_HB_P2-S_HydroBreaklines_NZTM<br>PRJ44793_HB_P2-S_LiDAR_Extent_NZTM<br>PRJ44793_HB_P2-S_OceanClip_NZTM<br>PRJ44793_HB_P2-S_TileIndex_NZTM<br>PRJ44793_HB_P2-S_Trajectories_NZTM | ESRI Shapefile format:<br>Hydro breaklines<br>Extent<br>Ocean clip boundary<br>Tile Index<br>Flight lines |
| Readme_PRJ44793_06.pdf | Metadata Report |
| PRJ44793_06_FileListing.txt | List of product files delivered in this volume |

# 2. METADATA

| Source Data | Source | Description | Ref No | Date |
|---|---|---|---|---|
| LiDAR & Imagery | Woolpert | TerrainMapper2 - 527 | FL020423 | 20.09.2023 |
| | | TerrainMapper2 - 527 | FL020511 | 01.10.2023 |
| | | TerrainMapper2 - 527 | FL020533 | 04.10.2023 |
| | | TerrainMapper2 - 527 | FL020539 | 04.10.2023 |
| | | TerrainMapper2 - 527 | FL020542 | 05.10.2023 |
| | | TerrainMapper2 - 527 | FL020547 | 05.10.2023 |
| | | TerrainMapper2 - 527 | FL020608 | 13.10.2023 |
| | | TerrainMapper2 – 527 | FL020770 | 04.11.2023 |
| | | TerrainMapper2 – 527 | FL020842 | 12.11.2023 |
| | | TerrainMapper2 – 527 | FL020852 | 12.11.2023 |
| | | TerrainMapper2 – 527 | FL020875 | 14.11.2023 |
| | | TerrainMapper2 – 527 | FL021135 | 15.12.2023 |
| | | TerrainMapper2 – 527 | FL021182 | 21.12.2023 |
| | | TerrainMapper2 – 527 | FL021325 | 08.01.2024 |
| | | TerrainMapper2 – 527 | FL021336 | 09.01.2024 |
| | | TerrainMapper2 – 527 | FL021416 | 19.01.2024 |
| | | TerrainMapper2 – 527 | FL021445 | 23.01.2024 |
| | | TerrainMapper2 – 527 | FL021459 | 24.01.2024 |

| | | | | |
|---|---|---|---|---|
| | | TerrainMapper2 – 527 | FL021470 | 26.01.2024 |
| | | TerrainMapper2 – 527 | FL021486 | 28.01.2024 |
| | | TerrainMapper2 – 527 | FL021588 | 08.02.2024 |
| | | TerrainMapper2 – 527 | FL021631 | 13.02.2024 |
| | | TerrainMapper2 – 527 | FL021640 | 14.02.2024 |
| | | TerrainMapper2 – 527 | FL021898 | 11.03.2024 |
| | | TerrainMapper2 – 527 | FL022244 | 19.04.2024 |
| GPS Base Data | GeoNET /LINZ | GeoNET CORS, LINZ GDB co-ordinates | As above | As above |
| Control | Sounds Surveying | RTK GNSS | PRJ44793 | 27/09/2023 – 8/10/2023 & 23-28/11/2023 |

| LiDAR Characteristics | Description |
|---|---|
| Format | LAZ 1.4 |
| Emitted Density | 8 ppm2 |
| Tile size | 480m x 720m (NZ Topo 50, 1:1000 tiles) |
| ICSM Classification | Level 2. Ground surface improvement |
| Capture Constraints | Coastline captured within 3 hours of low tide. |

| Number | Point Class | Description | ICSM | CI % |
|---|---|---|---|---|
| 1 | Default | Unclassified | 1 | 95 |
| 2 | Ground | Ground | 2 | 98 |
| 3 | Low vegetation | < 2.0 m | 1 | 95 |
| 4 | Medium vegetation | 2.0 – 8.0 m | 1 | 95 |
| 5 | High vegetation | > 8.0 m | 1 | 95 |
| 6 | Buildings, structures | Buildings, houses, sheds, silos etc. | 1 | 95 |
| 7 | Low / high points | Spurious low point returns (unusable) | 2 | 98 |
| 9 | Water | Any point in water | 2 | 98 |
| 17 | Bridge | Any bridge or overpass | 2 | 98 |
| 18 | High Noise | Spurious high point returns (unusable) | 2 | 98 |

| Ortho Characteristics | Description |
|---|---|
| Format | GeoTIFF |
| Ground Sample Distance | 12.5cm |
| Terrain Model | Concurrent LiDAR |
| Tile size | 480m x 720m (NZ Topo 50, 1:1000 tiles) |
| Sample Type | 8-bit Integer |
| Image Bands | RGBI |
| Orientation/AT | Aero Triangulation/GPS-IMU |
| Capture Constraints | Nil specific to imagery – captured under LiDAR capture conditions above, in daylight hours |

| Reference Systems | Horizontal | Vertical |
|---|---|---|
| Datum | NZGD2000 | NZVD2016 |
| Projection | NZTM2000 | N/A |
| Geoid Model | N/A | NZGeoid2016 |

| Accuracy Specification | Measured Point | Derived Point | Basis of Estimation |
|---|---|---|---|
| Control Points | 0.05m | | Survey Methodology |
| LiDAR (Horizontal) | 0.40m | | Project design |
| LiDAR (Vertical) | 0.10m | | Project design |
| Orthophoto | 0.30m | | Project design |

## Notes On Expected Accuracy

- Values shown represent standard error (68% confidence level or 1 sigma), in metres
- “Derived points” are those interpolated from a terrain model.
- “Measured points” are those observed directly.
- Accuracy estimates for terrain modeling by LiDAR refer to the terrain definition on clear ground.
- LASER strikes have been classified into “ground” and “non-ground”, based upon algorithms tailored for major terrain/vegetation combinations existing in the project area. The definition of the ground may be less accurate in isolated pockets of dissimilar terrain/vegetation combinations.

## Limitations of Data

- The definition of the ground under trees may be less accurate.
- Buildings and vegetation have been classified automatically: ISCM Level 1, 95% confidence. These classes may contain artefacts such as powerlines, car's, containers etc. Be aware when using the LAZ and CHM products.

## Data Validation – LiDAR Data

- Vertical Accuracy Validation - Ground data in this volume has been compared to ~1636 test points obtained by field survey and assumed to be error-free. The test points were distributed in 33 groups across the mapping area and located on clear ground. Comparison of the test points with elevations interpolated from measured data resulted in:

  Mean difference: 0.040 m

  Std Deviation: 0.036 m

  Standard Error (RMS): 0.053 m or 0.104 m @ 95% CI

  DEM and control comparisons were made to the adjacent Priority One block to ensure consistency, these align within 0.04m.

  A shift of -0.076 was applied to the data before testing against the check points to produce the results shown above.

- Horizontal Accuracy – the LiDAR data was compared to visible horizontal control using the intensity imagery. The data was found to fit well. The expected horizontal accuracy at 95% CI is 38cm, within the specification of 100cm.
- Data classification has been manually checked and edited against available imagery.
- More information is available in the Survey Report: PRJ44793_05-07_HawkesBay_P2_LiDAR_Report.pdf

## Data Validation – Orthophoto Image

- This data has not been field tested for accuracy. Spot checks comparing the orthophoto with horizontal check points supported the estimated horizontal accuracy of < 0.300 m. Full proof of accuracy achieved requires comparison to independent test points.

# 3. CONDITIONS OF SUPPLY

The data in this volume has been commissioned by the National Institute of water and Atmospheric Research Limited **NIWA**.

The data in this volume is provided by Woolpert to **NIWA** under the Terms of Engagement described in the **NIWA Woolpert Imagery Contract Agreement, dated 16th August 2023**. Which provides the client, **NIWA** (or its nominee) with ownership of the deliverables to fully protect, realise and use all delivered data and reports, allowing release of data for re-use with a Creative Commons license (CC BY) with attribution to the buyer in line with the New Zealand Government Open Access Licensing framework (NZGOAL). This specification places no restrictions on the rights of Woolpert to resell data or derivative products.

1. This file (Readme_PRJ44793_06.pdf) will always be stored with the unaltered data contained in this volume.

This data is provided in accordance with the specifications agreed with NIWA and Land Information NZ. Any problems associated with the information in the data files contained in this volume should be reported to Woolpert, Asia-Pacific.

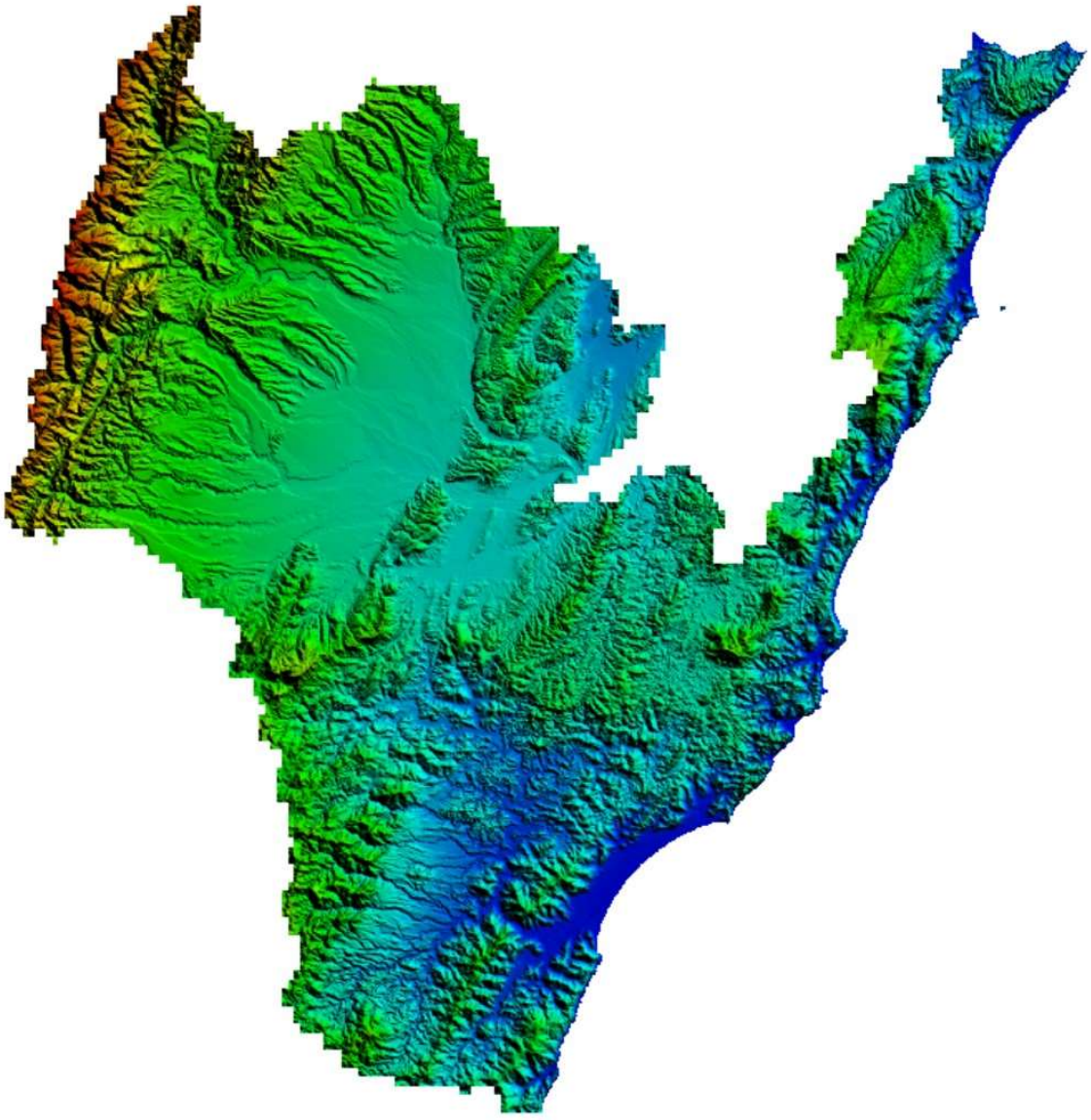
Woolpert NZ Limited
Level 1
6 Ossian St
Napier 4110
New Zealand

woolpert.com
Twitter Facebook LinkedIn Instagram

# 4. VALIDATION

PRJ44793_06 – P2-S

Colour Elevation



# NIWA

# POST CYCLONE GABRIELLE HAWKES BAY LiDAR

# VOLUME: PRJ44793_07

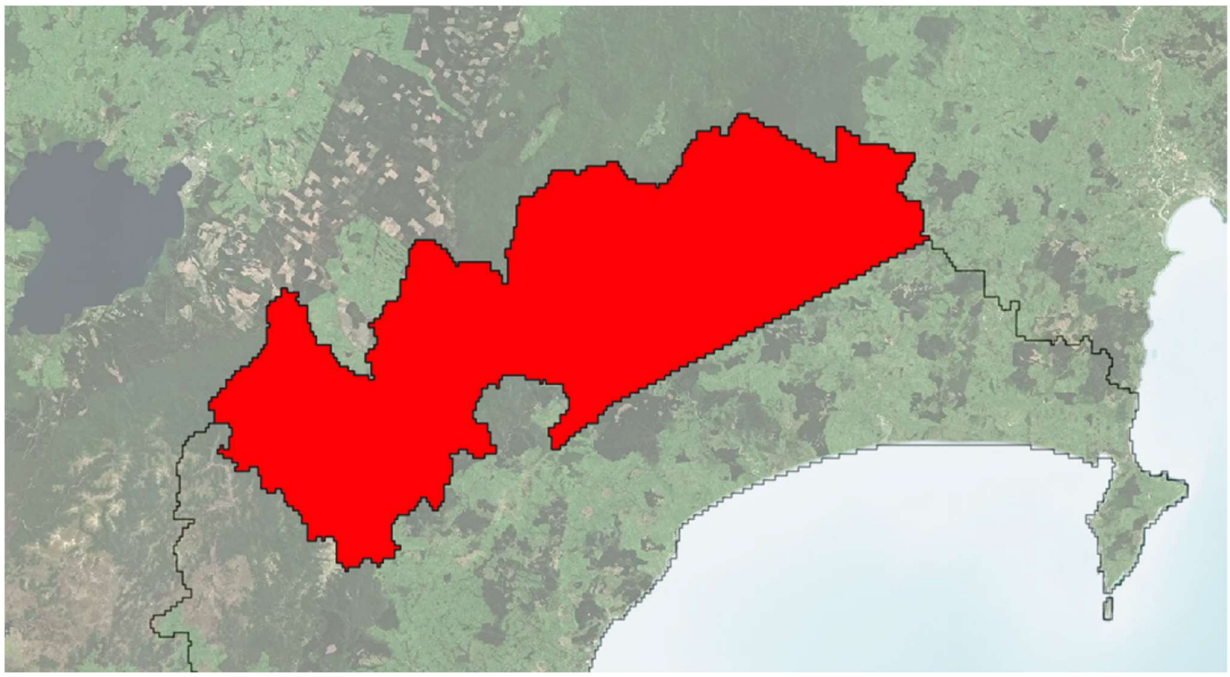
## PROJECT SUMMARY

The National Institute of Water and Atmospheric Research Limited (NIWA) contracted Woolpert to capture and process LiDAR and concurrent imagery over Hawkes Bay, post Cyclone Gabrielle.

This report is for the products supplied in Northern Hawkes Bay for Priority Two - volume PRJ44793_07, approximately 3,581 $km^2$.

This area was captured between the 21st of September 2023 – 27th of April 2024.

The survey was planned to achieve ≤ 20cm vertical accuracy (95% CI), ≤ 100cm horizontal accuracy (95% CI) with an emitted pulse density of 8ppsm, and ground classification to ICSM level 2.



# DATA SUMMARY

This volume includes the following data in NZTM2000 projection, NZVD2016 vertical datum:

- 10,361 x NZTopo50 1:1000 tiles are included in PRJ44793_07
- 10,361 tiles of ICSM L2 Colourised Classified Point Cloud data in LAZ v1.4 format
- 10,361 tiles of 1m cell DEM in GeoTIFF format
- 10,361 tiles of 1m cell DSM in GeoTIFF format
- 10,361 tiles of 1m cell CHM in GeoTIFF format
- 10,361 tiles of 12.5cm gsd RGBI Imagery in GeoTIFF format
- Ancillary files in ESRI Shapefile format – Tile Index, Delivery Extent, Flightlines, Hydro Breaklines.
- File listing in text file format
- Metadata file: This document in PDF format

# CONTENTS

1. Data Information ..... 4
2. Metadata ..... 5
3. Conditions Of Supply ..... 11
4. Validation ..... 12

# 1. DATA INFORMATION

Data supply: LINZ AWS
Number of files: 93,265 data files, 1 file list, 1 survey report, 1 metadata file
Data formatted on: 31/07/2024- 28/08/2024
Metadata Document: This file

| Previous Deliveries | Date | Title | Contents |
|---|---|---|---|
| PRJ44793_01 | 22.12.2023 | Post Cyclone Gabrielle Hawkes Bay LiDAR | Sample of deliverables from P1-SE |
| PRJ44793_02 | 08.04.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P1-SE |
| PRJ44793_03 | 05.04.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P1-N |
| PRJ44793_04 | 06.04.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P1-W |
| PRJ44793_04-02 | 14-22.05.2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | Resubmissions for Priority 1 blocks SE, N, W |
| PRJ44793_05 | 12-28/05/2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P2-NE |
| PRJ44793_06 | 19/06/2024 – 17/07/2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | P2-S |
| PRJ44793_05 | 13&23/08/2024 | Post Cyclone Gabrielle Hawkes Bay LiDAR | Resubmissions P2-NE |

| File Naming in this Delivery | Contents |
|---|---|
| CL2_BG38_2023_1000_3248.laz | Colourised classified point cloud in LAZ v1.4 format |
| DEM_ BG38_2023_1000_3248.tif/tfw | 1m grid DEM |
| DSM_ BG38_2023_1000_3248.tif/tfw | 1m grid DSM |
| CHM_ BG38_2023_1000_3248.tif/tfw | 1m grid CHM (canopy height above ground) |
| RGBI_ BG38_2023_1000_3248.tif/tfw | 12.5cm gsd RGBI imagery |
| Ancillary files:<br>PRJ44793_HB_P2-N_HydroBreaklines_NZTM<br>PRJ44793_HB_P2-N_ProjectExtent_NZTM<br>PRJ44793_HB_P2-N_TileIndex_NZTM<br>PRJ44793_HB_P2-N_Trajectories_NZTM | ESRI Shapefile format:<br>Volume 07 Hydro breaklines<br>Volume 07 Extent<br>Volume 07 Tile Index<br>Volume 07 Flight lines |
| Readme_PRJ44793_07.pdf | Metadata Report |
| PRJ44793_07_FileListing.txt | List of product files delivered in this volume |

# 2. METADATA

| Source Data | Source | Description | Ref No | Date |
|---|---|---|---|---|
| LiDAR & Imagery | Woolpert | TerrainMapper2 - 527 | FL020424 | 21.09.2023 |
| | | TerrainMapper2 - 527 | FL020430 | 21.09.2023 |
| | | TerrainMapper2 – 527 | FL020601 | 12.10.2023 |
| | | TerrainMapper2 – 527 | FL020655 | 20.10.2023 |
| | | TerrainMapper2 – 527 | FL020842 | 12.11.2023 |
| | | TerrainMapper2 – 527 | FL020856 | 13.11.2023 |
| | | TerrainMapper2 – 527 | FL020865 | 13.11.2023 |
| | | TerrainMapper2 – 527 | FL020875 | 14.11.2023 |
| | | TerrainMapper2 – 527 | FL021060 | 07.12.2023 |
| | | TerrainMapper2 – 527 | FL021135 | 15.12.2023 |

| | | | | |
|---|---|---|---|---|
| | | TerrainMapper2 – 527 | FL021141 | 16.12.2023 |
| | | TerrainMapper2 – 527 | FL021150 | 16.12.2023 |
| | | TerrainMapper2 – 527 | FL021156 | 17.12.2023 |
| | | TerrainMapper2 – 527 | FL021159 | 17.12.2023 |
| | | TerrainMapper2 – 527 | FL021182 | 21/12/2023 |
| | | TerrainMapper2 – 527 | FL021358 | 12/01/2024 |
| | | TerrainMapper2 – 527 | FL021359 | 12/01/2024 |
| | | TerrainMapper2 – 527 | FL021459 | 24/01/2024 |
| | | TerrainMapper2 – 527 | FL021581 | 7/02/2024 |
| | | TerrainMapper2 – 527 | FL021627 | 12/02/2024 |
| | | TerrainMapper2 – 527 | FL021860 | 8/03/2024 |
| | | TerrainMapper2 – 527 | FL021893 | 11/03/2024 |
| | | TerrainMapper2 – 527 | FL021898 | 11/03/2024 |
| | | TerrainMapper2 – 527 | FL022290 | 23/04/2024 |
| | | TerrainMapper2 – 527 | FL022293 | 24/04/2024 |
| | | TerrainMapper2 – 527 | FL022321 | 27/04/2024 |
| GPS Base Data | GeoNET /LINZ | GeoNET CORS, LINZ GDB co-ordinates | As above | As above |
| Control | Sounds Surveying | RTK GNSS | PRJ44793 | 27/09/2023 – 8/10/2023 & 23-28/11/2023 |

| LiDAR Characteristics | Description |
|---|---|
| Format | LAZ 1.4 |
| Emitted Density | 8 ppm2 |
| Tile size | 480m x 720m (NZ Topo 50, 1:1000 tiles) |
| ICSM Classification | Level 2. Ground surface improvement |
| Capture Constraints | Coastline captured within 3 hours of low tide. |

| Number | Point Class | Description | ICSM | CI % |
|---|---|---|---|---|
| 1 | Default | Unclassified | 1 | 95 |
| 2 | Ground | Ground | 2 | 98 |
| 3 | Low vegetation | < 2.0 m | 1 | 95 |
| 4 | Medium vegetation | 2.0 – 8.0 m | 1 | 95 |
| 5 | High vegetation | > 8.0 m | 1 | 95 |
| 6 | Buildings, structures | Buildings, houses, sheds, silos etc. | 1 | 95 |
| 7 | Low Noise | Spurious low point returns (unusable) | 2 | 98 |
| 9 | Water | Any point in water | 2 | 98 |
| 17 | Bridge | Any bridge or overpass | 2 | 98 |
| 18 | High Noise | Spurious high point returns (unusable) | 2 | 98 |

| Ortho Characteristics | Description |
|---|---|
| Format | GeoTIFF |
| Ground Sample Distance | 12.5cm |
| Terrain Model | Concurrent LiDAR |
| Tile size | 480m x 720m (NZ Topo 50, 1:1000 tiles) |
| Sample Type | 8-bit Integer |
| Image Bands | RGBI |
| Orientation/AT | Aero Triangulation/GPS-IMU |
| Capture Constraints | Nil specific to imagery – captured under LiDAR capture conditions above, in daylight hours |

| Reference Systems | Horizontal | Vertical |
|---|---|---|
| Datum | NZGD2000 | NZVD2016 |
| Projection | NZTM2000 | N/A |
| Geoid Model | N/A | NZGeoid2016 |

| Accuracy Specification | Measured Point | Derived Point | Basis of Estimation |
|---|---|---|---|
| Control Points | 0.05m | | Survey Methodology |
| LiDAR (Horizontal) | 0.40m | | Project design |
| LiDAR (Vertical) | 0.10m | | Project design |
| Orthophoto | 0.30m | | Project design |

## Notes On Expected Accuracy

- Values shown represent standard error (68% confidence level or 1 sigma), in metres
- “Derived points” are those interpolated from a terrain model.
- “Measured points” are those observed directly.
- Accuracy estimates for terrain modeling by LiDAR refer to the terrain definition on clear ground.
- LASER strikes have been classified into “ground” and “non-ground”, based upon algorithms tailored for major terrain/vegetation combinations existing in the project area. The definition of the ground may be less accurate in isolated pockets of dissimilar terrain/vegetation combinations.

## Limitations of Data

- The definition of the ground under trees may be less accurate.
- Buildings and vegetation have been classified automatically: ISCM Level 1, 95% confidence. These classes may contain artefacts such as powerlines, car’s, containers etc. Be aware when using the LAZ and CHM products.

## Data Validation – LiDAR Data

- Vertical Accuracy Validation - Ground data in this volume has been compared to ~1043 test points obtained by field survey and assumed to be error-free. The test points were distributed in 19 groups across the mapping area and located on clear ground. Comparison of the test points with elevations interpolated from measured data resulted in:

  Mean difference: 0.024 m

  Std Deviation: 0.058 m

  Standard Error (RMS): 0.063 m or 0.124 m @ 95% CI

  Comparison by site:

| Site | Post dZ | Post RMS | Post SD |
|---|---|---|---|
| *Average* | *0.044* | | |
| 1028 | 0.128 | 0.130 | 0.023 |
| 1029 | 0.008 | 0.022 | 0.021 |
| 1049 | -0.005 | 0.019 | 0.018 |
| 1050 | 0.085 | 0.087 | 0.020 |
| 1056 | -0.004 | 0.013 | 0.012 |
| 1057 | 0.053 | 0.055 | 0.015 |

P1 N | P2 NW | P2 NN

| Site | Post dZ | Post RMS | Post SD |
|---|---|---|---|
| *Average* | *0.014* | | |
| 1004 | 0.063 | 0.067 | 0.024 |
| 1007 | -0.006 | 0.020 | 0.020 |
| 1008 | 0.081 | 0.085 | 0.029 |
| 1009 | -0.087 | 0.089 | 0.020 |
| 1010 | 0.022 | 0.027 | 0.017 |
| 1011 | 0.020 | 0.031 | 0.024 |
| 1012 | 0.045 | 0.049 | 0.021 |
| 1013 | -0.031 | 0.036 | 0.019 |
| 1016 | -0.025 | 0.033 | 0.022 |
| 1017 | 0.064 | 0.072 | 0.033 |
| 1027 | 0.064 | 0.071 | 0.032 |
| 1030 | 0.030 | 0.035 | 0.018 |
| 1123 | -0.059 | 0.062 | 0.019 |

P1 N | P2 NN | P2 NW

  This volume is comprised for 2 blocks named P2-NW and P2-NN. DZ shifts were applied to each block following comparison to control and comparisons to the adjoining Priority One and Priority Two blocks.

  Shifts applied were: -0.098m (P2-NW shift to P1-W and P1-N)

  -0.083m (P2-NN shift to P2-NW, P2-NE and P1-N)

  These vertical shifts were applied to the data prior to comparison to the test points shown above.

- Horizontal Accuracy – the LiDAR data was compared to visible horizontal control using the intensity imagery. The data was found to fit well. The expected horizontal accuracy at 95% CI is 38cm, within the specification of 100cm.
- Data classification has been manually checked and edited against available imagery.

## Data Validation – Orthophoto Image

- This data has not been field tested for accuracy. Spot checks comparing the orthophoto with horizontal check points supported the estimated horizontal accuracy of < 0.300 m. Full proof of accuracy achieved requires comparison to independent test points.

# 3. CONDITIONS OF SUPPLY

The data in this volume has been commissioned by the National Institute of water and Atmospheric Research Limited **NIWA**.

The data in this volume is provided by Woolpert to **NIWA** under the Terms of Engagement described in the **NIWA Woolpert Imagery Contract Agreement, dated 16th August 2023**. Which provides the client, **NIWA** (or its nominee) with ownership of the deliverables to fully protect, realise and use all delivered data and reports, allowing release of data for re-use with a Creative Commons license (CC BY) with attribution to the buyer in line with the New Zealand Government Open Access Licensing framework (NZGOAL). This specification places no restrictions on the rights of Woolpert to resell data or derivative products.

1. This file (Readme_PRJ44793_07.pdf) will always be stored with the unaltered data contained in this volume.

This data is provided in accordance with the specifications agreed with NIWA and Land Information NZ. Any problems associated with the information in the data files contained in this volume should be reported to Woolpert, Asia-Pacific.

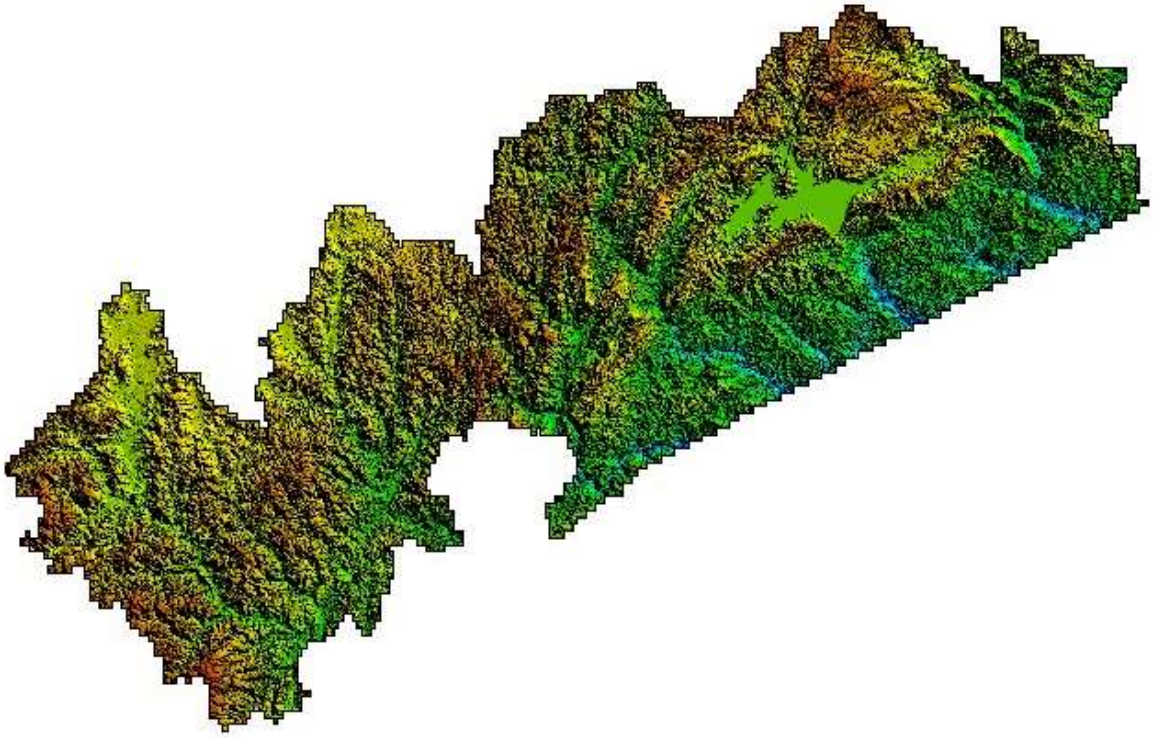
Woolpert NZ Limited
Level 1
6 Ossian St
Napier 4110
New Zealand

woolpert.com
Twitter Facebook LinkedIn Instagram

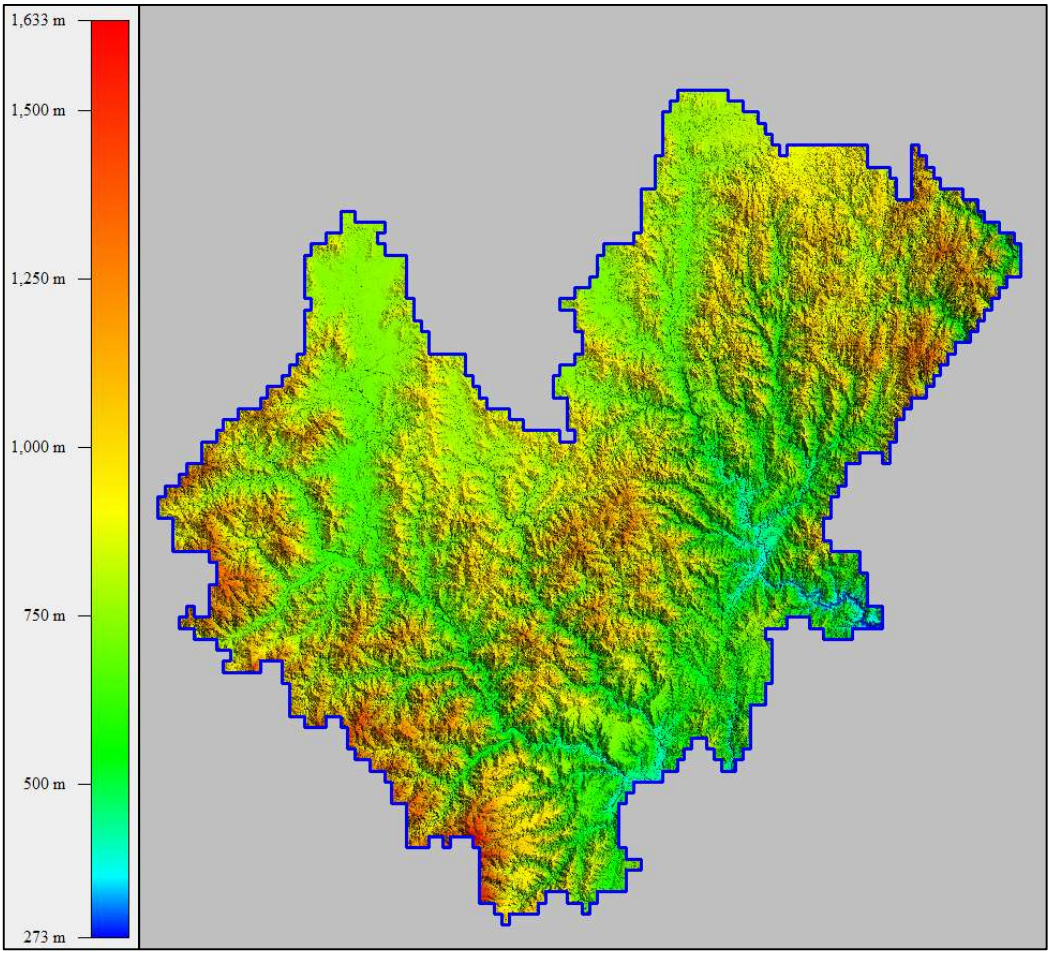
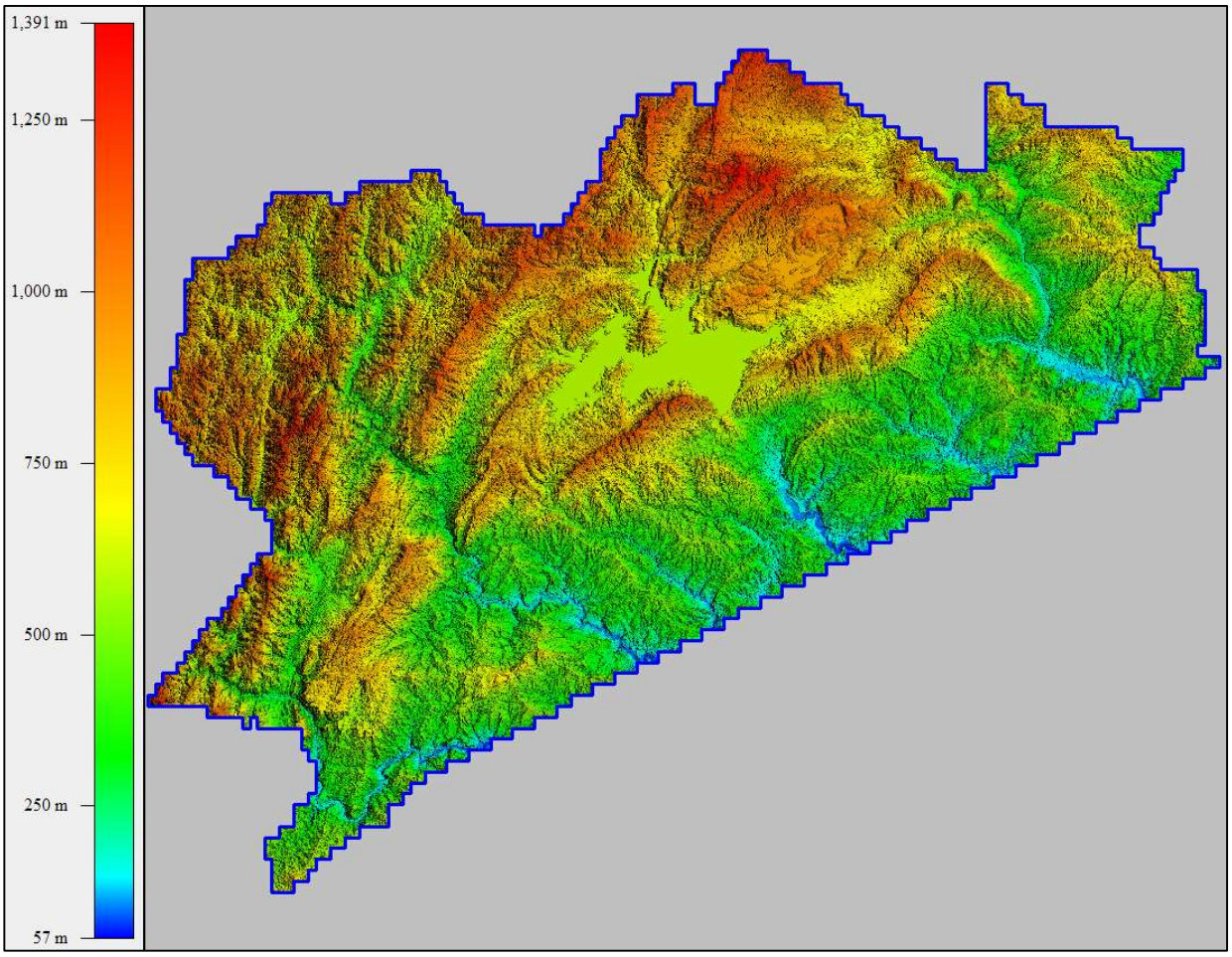
# 4. VALIDATION

PRJ44793_07 – P2-N

Colour Elevation Overview



P2-NW



P2-NN